Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	0157CY241012
Project Name	EV Market Analysis (Team: Data Warrior)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python, PowerShell, HTML, CSS, JavaScript	Used to generate and validate data, and to build the web dashboard.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV files (ev_market_data.csv, cleaned CSV)	Simple and portable; enough for 1,200 EV records.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Data Connection (CSV extract)	Loads the cleaned CSV directly into Tableau with no extra tools.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop calculated fields, PowerShell scripts	Cleans data and calculates price per km, revenue and CO2 saved.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Public, Chart.js	Builds interactive charts, KPIs and the Tableau Story.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public, HTML5/CSS3 web app, Flask, GitHub Pages	Lets anyone open and filter the dashboard in a browser.